Stefano Stucchi

Curriculum Vitae

Bergamo, Italy
Mechanical Engineering PhD Candidate
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Education

2026–Present : **Ph.D., Mechanical Engineering**, *Politecnico di Milano*, Milano Bovisa (MI).
Investigation of gait-derived digital biomarkers of Alzheimer's disease and cognitive decline, and development of wearable and exoskeleton-based technologies for assessment, rehabilitation, and home-based assistance.

2023–2026 : **MSc, Sports Engineering - Mechanical Engineering**, *Politecnico di Milano*, Lecco (LC), Grade: 110/110 with honours.
Advanced exposure to various research aspects related to sports and Paralympics, with specific focus on sensors and technologies adopted in these fields.

2020–2023 : **Bachelor of Engineering, Aerospace Engineering**, *Politecnico di Milano*, Milano Bovisa (MI), Grade: 103/110.
Advanced exposure to various areas of mechanics, aeronautics and space science.

2015–2020 : **High School Diploma, Scientific focus**, *Liceo Scientifico "Filippo Lussana"*, Bergamo (BG).

Master Thesis

August,2023–March,2026 : ***Methodological Contributions to Functional and Technical Assessment in Paralympic Water Polo Classification***, *Authors: Pietrobon Alberto, Stucchi Stefano.*

Supervisor : **Dr. Marta Gandolla**, *Associate Professor, Department of Mechanical Engineering (DMEC)*, Politecnico di Milano.

Main Contributions:

- Developed a **theoretical protocol** for instrumented dry-land assessments (strength, ROM, coordination) based on a comprehensive literature review and stakeholder interviews.
- Conducted an **experimental phase** validating in-water technical tests, utilizing statistical models to analyze the correlation between impairment, training volume, and athletic performance.
- Established a quantitative methodology to replace manual ordinal scales, enhancing objectivity, repeatability, and evidence-based standards in Paralympic water polo classification.

Engineering Projects

MSc, Sport Engineering - Mechanical Engineering, Politecnico di Milano

June,2025 ***ReActive - smart pants for functional electrical stimulation (FES) and EMG monitoring***, *Authors: Comotti Leonardo, Pietrobon Alberto, Stucchi Stefano.*
Design and prototyping of a smart garment integrating EMG and FES technologies for elderly rehabilitation, utilizing Autodesk Inventor for digital mapping and specialized conductive materials to ensure stretch resistance and low impedance.

Advisor : **Dr. Giuseppe Andreoni**, *Associate Professor, Department of Design (DESIS)*, Politecnico di Milano.

June,2025 ***Paralympics and Sport Rehabilitation - Sport for inclusion and active ageing***, *Authors: Baronchelli Greta, Bianucci Allegra, Comotti Leonardo, Pietrobon Alberto, Stucchi Stefano.*

Development of mechanical and ergonomic enhancements for PoliMi's Functional Electrical Stimulation (FES) trike, implementing removable assist handles, dynamic armrests, and a car-style reclining seat mechanism to improve accessibility and visibility for active ageing. Testing performed through structural FEA (safety factor 2.1) and field-of-view analysis.

Advisor : **Dr. Marta Gandolla & Dr. Emilia Ambrosini**, *Associate Professors, Department of Mechanical Engineering (DMEC) & Department of Electronics, Information and Bioengineering (DEIB)*, Politecnico di Milano.

January,2025 ***VRocky: A VR boxing simulator for amateurs***, *Authors: Airoldi Lorenzo, Biffi Claudio, Stucchi Stefano.*

Development of an interactive VR boxing simulator using Unity and Captury Live motion capture to provide a safe, injury-free training environment for amateurs. Implementation of "learn-by-doing" tutorials and four specialized mini-games focusing on motor skills, coordination, and reactive training, and validation of the system usability through a SUS (System Usability Scale) assessment.

Advisor : **Dr. Mario Covarrubias**, *Associate Professor, Department of Mechanical Engineering (DMEC)*, Politecnico di Milano.

June,2024 ***Kinematic and dynamic validation of OpenCap motion capture system through Xsens IMUs***, *Authors: Biffi Claudio, Comotti Leonardo, Pietrobon Alberto, Stucchi Stefano, Toniolo Filippo.*

Validation of the OpenCap motion capture system against Xsens inertial sensors, demonstrating comparable accuracy in knee/ankle kinematics and jump height estimation through GRF profiles and peak power analysis.

Advisor : **Dr. Manuela Galli**, *Associate Professor, Department of Electronics, Information and Bioengineering (DEIB)*, Politecnico di Milano.

Publications

In Conference Proceedings

2026 **Stefano Stucchi**, Giuseppe Andreoni, Denise Bentivoglio, Leonardo Comotti, and Alberto Pietrobon. Seamless integration of fes and emg/imu in smart pants to enhance autonomy in elderly fes-bike rehabilitation. In *2026 Soft Robotics for Healthcare (SoRoH) - Bristol (UK)*. SoRoH, 2026.

2025 **Stefano Stucchi**, Lorenzo Airoldi, Claudio Biffi, Mario Covarrubias, and Sara Arlati. Vrocky: A vr boxing simulator for amateurs and young students with disabilities. ***CAD'25***, pages 284–289. CAD'25, 2025.

Technical Skills

Programming Languages	MATLAB, Python - for Data Science & Engineering (NumPy, Pandas, SciPy, Matplotlib)
CAD Software	SolidWorks, Autodesk Inventor
Tools & Frameworks	Jupyter, Visual Studio Code, Figma
Documentation & Publishing	LaTeX
Productivity Tools	Microsoft Office (Excel, Word, PowerPoint)

Languages

Italian	Native
English	Advanced C1 - IELTS Certified (2023)